

Project Interim Report
On
Weather Monitoring System
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

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Abstract

The Weather Monitoring System is a project designed to measure and display real-time weather conditions such as temperature, humidity, and atmospheric pressure using sensors and a digital interface.

This system is needed because accurate and timely weather information is important for agriculture, travel, and daily planning. Traditional methods are often slow and less reliable, while an automated system provides faster and more precise data.

The system works by collecting data through sensors, processing it using a microcontroller or software, and displaying the results on a screen or application. It can also store past data for analysis and forecasting. Overall, the Weather Monitoring System helps users make better decisions by providing quick and reliable weather updates.

Introduction

The Weather Monitoring System is a project that helps us observe and understand weather conditions in real time. It collects data like temperature, humidity, and air pressure using sensors and shows it in a simple and easy-to-read format.

This project is useful because traditional ways of checking weather can be slow and sometimes inaccurate. With this system, we can get quick and reliable updates, which can help in daily planning as well as in fields like agriculture and travel.

The system works by taking input from sensors, processing it through a program or microcontroller, and displaying the results on a screen. It can also store past data, which can be helpful for analysis and future predictions.

Objectives of the Project:

- Monitor weather conditions in real time
- Store and analyze collected data
- Reduce manual effort and errors
- Provide fast and accurate updates
- Use OOP concepts for better structure and design

Scope

The scope of the Weather Monitoring System includes real-time tracking of environmental conditions for personal, agricultural, and industrial use. It can be expanded to provide weather forecasting and alerts for extreme conditions. The system can also be integrated with mobile or web applications for remote access. In the future, it can support smart city and IoT-based environmental monitoring solutions.

System Requirements

Hardware Requirements:

- Temperature and humidity sensors (e.g., DHT11/DHT22)
- Microcontroller (e.g., Arduino or similar)
- Display unit (LCD/LED or computer screen)
- Power supply
- Connecting wires and basic circuit components

Software Requirements:

- Programming language (e.g., C++, Java, or Python)
- IDE (e.g., Arduino IDE or any suitable compiler)
- Operating System (Windows/Linux/Mac)
- Optional: Database for storing historical data

Other Requirements:

- Basic knowledge of programming and electronics
- Internet connection (if using cloud or web-based monitoring)

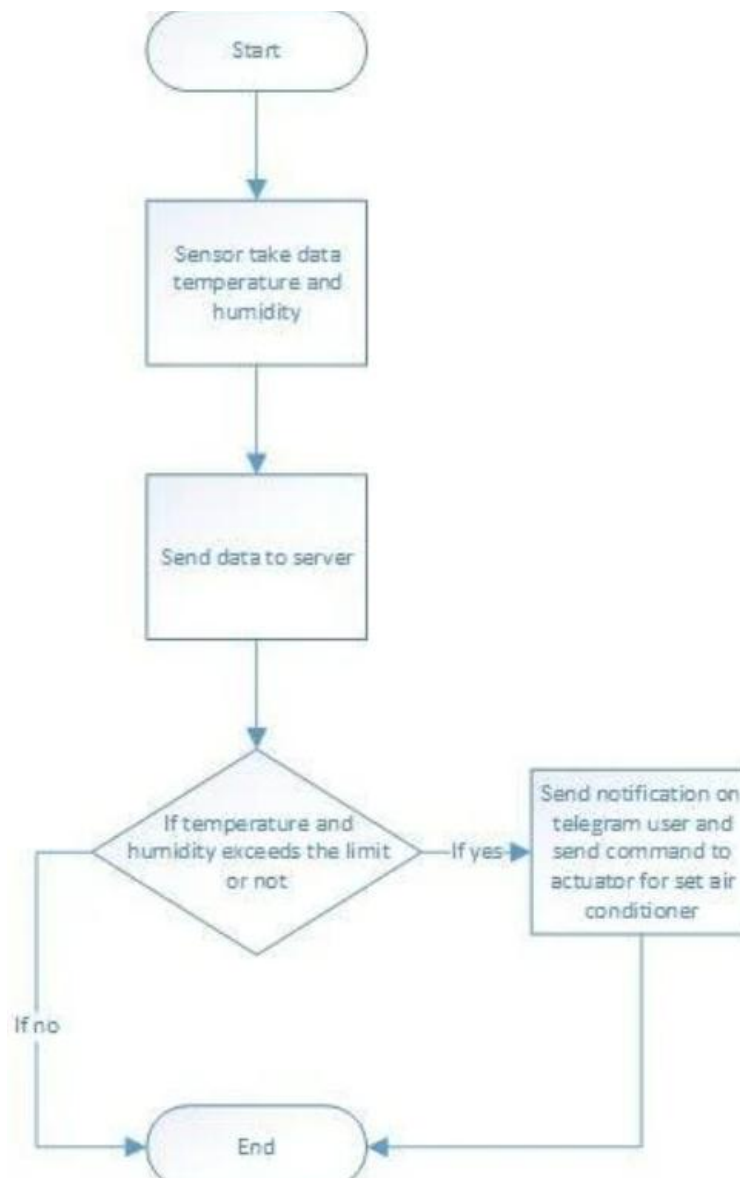
Methodology

Methodology (Working of the Project)

The working of the Weather Monitoring System follows a simple step-by-step flow:

1. **Data Collection:**
First, the system collects weather data such as temperature, humidity, and pressure using sensors.
2. **Data Processing:**
The collected data is then sent to a microcontroller or program, where it is processed and converted into meaningful values.
3. **Display of Data:**
After processing, the system shows the current weather details on a screen or application in an easy-to-understand format.
4. **Data Storage:**
The system stores the collected data so it can be used later for analysis or comparison.
5. **Update & Monitoring:**
The data keeps updating continuously, allowing users to monitor changes in real time.
6. **Optional Alerts/Analysis:**
The system can also give alerts (like high temperature or humidity) and help in basic future predictions based on stored data.

Flowchart



Flowchart Steps –

1. Start
2. Sensor takes data (temperature and humidity)
3. Send data to server
4. Check if temperature and humidity exceed the limit
5. If YES:
 - Send notification to user (Telegram)
 - Send command to actuator (e.g., turn on air conditioner)
 - End
6. If NO:
 - End

ClassName

- o Class Name: WeatherData
- o Variables:
 - float temperature
 - float humidity
 - float pressure
 - float dataHistory[] // array to store past weather records
- o Methods:
 - storeData()
 - getData()

Implementation

The project is designed using different classes, each handling a specific task to keep the system organized.

- **Sensor Class:**
This class is responsible for collecting real-time data such as temperature and humidity from the environment.
- **WeatherData Class:**
It stores the collected data and manages past records using an array. This helps in analyzing weather patterns over time.
- **Controller Class:**
This acts as the main logic handler. It takes data from the sensor, checks conditions (like limit exceed), and decides what action to perform.
- **Display Class:**
It shows the processed data to the user in a simple and readable format. Overall, the system works by collecting data → processing it → storing it → displaying results.

Sample Input / Output

Sample Input:

```
Temperature = 35°C  
Humidity = 80%
```

Sample Output:

```
Current Weather:  
Temperature: 35°C  
Humidity: 80%  
  
Alert: Limit Exceeded!  
Action: Notification sent & AC turned ON
```

Future Enhancements

- Add real-time mobile app integration
- Improve accuracy with advanced sensors
- Include weather prediction using AI/ML
- Store data on cloud for remote access
- Add alert system for multiple conditions (rain, storm, etc.)

Gantt Chart

1. Planning & Structure

A Gantt chart helps in breaking the project into smaller tasks such as design, coding, and testing, and assigning timelines to each of them. Even if the project is already built, rebuilding or modifying it still requires proper planning to ensure smooth execution.

2. Time Management

It clearly shows when each task starts and when it should be completed. This helps in managing time effectively, avoiding delays, and reducing last-minute pressure.

Conclusion

The Weather Monitoring System is a simple and useful project that helps in collecting and displaying real-time weather data efficiently. It reduces manual effort and provides accurate and quick updates, making it helpful for daily use as well as for applications like agriculture and travel. By using a structured approach and OOP concepts, the system becomes easy to manage and extend. Overall, this project demonstrates how technology can be used to monitor environmental conditions in a reliable and effective way.

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